

Linear algebra and geometry 1 - TTPU

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Grade 13.20 out of 30.00 (44%)

Question 1

Correct

Mark 100.00 out of 100.00

Let $\vec{i}, \vec{j}, \vec{k}$ the versors of the coordinate axes of $\mathbb{R}^3$. For every $a \in \mathbb{R}$, let us consider the vectors

$$\vec{u} = a\vec{j} + a\vec{k}, \quad \vec{v} = 2\vec{i} + \vec{k}, \quad \vec{w} = \vec{k}.$$

Which of the following statements is true?

- ☒ (a) There exists a unique $a \in \mathbb{R}$ such that the vectors $\vec{u}, \vec{v}, \vec{w}$ are coplanar. ✓
- ☐ (b) For every $a \in \mathbb{R}$ the vectors $\vec{u}, \vec{v}, \vec{w}$ are coplanar.
- ☐ (c) For every $a \in \mathbb{R}$ the vectors $\vec{u}, \vec{v}, \vec{w}$ are linearly independent.
- ☐ (d) There exists a unique $a \in \mathbb{R}$ such that the vectors $\vec{u}, \vec{v}, \vec{w}$ are linearly independent.

Risposta corretta.

The correct answer is: There exists a unique $a \in \mathbb{R}$ such that the vectors $\vec{u}, \vec{v}, \vec{w}$ are coplanar.

Question 2

Incorrect

Mark -15.00 out of
100.00Given the plane $\pi : 3x - 5y - z = 0$ and the straight line

$$r : \begin{cases} 2x + y - z = 1 \\ 3x + 2y + z = 1. \end{cases}, \text{ find the correct statement.}$$

-
- ☒ (a) $r \subseteq \pi$. ✖
- ☐ (b) r and π are orthogonal.
- ☐ (c) None of the other statements is correct.
- ☐ (d) $r \cap \pi = \emptyset$.

Risposta errata.

The correct answer is: None of the other statements is correct.

Question 3

Correct

Mark 100.00 out of
100.00If the linear system $AX = B$ with 4 equations and 6 variables has ∞^m solutions with $m \leq 4$, then necessarily

-
- ☒ (a) $\text{rank}(A) \geq 2$. ✔
- ☐ (b) $\text{rank}(A) \leq 2$.
- ☐ (c) $\text{rank}(A|B) = 2$.
- ☐ (d) $\text{rank}(A) = 2$.

Risposta corretta.

The correct answer is: $\text{rank}(A) \geq 2$.

Question 4

Not answered

Marked out of
100.00

Let $\cdot$ denote the usual matrix multiplication. If

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 5 \\ 0 & 0 \end{bmatrix},$$

then

- ☐ (a) $\det(\mathbf{A} \cdot \mathbf{B}) = \det(\mathbf{A}) \cdot \det(\mathbf{B})$.
- ☐ (b) $\det(\mathbf{A} \cdot \mathbf{B}) = 0$.
- ☐ (c) $\det(\mathbf{A} \cdot \mathbf{B}) = 5$.
- ☐ (d) $\det(\mathbf{A} \cdot \mathbf{B}) = \det(\mathbf{B}) \cdot \det(\mathbf{A})$.

Risposta errata.

The correct answer is: $\det(\mathbf{A} \cdot \mathbf{B}) = 5$.

Question 5

Incorrect

Mark -15.00 out of
100.00

Let $f : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ be an endomorphism with characteristic polynomial $P(t) = t(t+3)(t^2-2)$.

Which one of the following statements is true?

- ☒ (a) $\dim(\text{Ker}(f)) = 2$. ✖
- ☐ (b) $\dim(\text{Im}(f)) = 1$.
- ☐ (c) $\dim(\text{Im}(f)) = 2$.
- ☐ (d) $\dim(\text{Im}(f)) = 3$.

Risposta errata.

The correct answer is: $\dim(\text{Im}(f)) = 3$.

Question 6

Correct

Mark 100.00 out of
100.00

In $\mathbb{R}^4$ let us consider the vector subspace

$$V = \{ (x, y, z, w) \mid x + y + z + w = 0, z - w = 0 \}.$$

Which one of the following statements is true?

- ☒ (a) V is the vector subspace generated by the vectors $(-2, 0, 1, 1)$ e $(1, -1, 0, 0)$. ✓
- ☐ (b) V is a subspace of dimension 1.
- ☐ (c) V is a line of $\mathbb{R}^4$.
- ☐ (d) V is the vector subspace generated by the vectors $(1, 1, 1, 1)$ e $(3, 1, 0, 0)$.

Risposta corretta.

The correct answer is: V is the vector subspace generated by the vectors $(-2, 0, 1, 1)$ e $(1, -1, 0, 0)$.

Question 7

Incorrect

Mark -15.00 out of
100.00

Let us consider the matrix $A = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 1 & h \\ 0 & h & 4 \end{pmatrix}$

with $h \in \mathbb{R}$.

Which one of the following statements is true?

- ☒ (a) A is diagonalizable only if $h = 0$. ✗
- ☐ (b) A is diagonalizable only if $h \neq 4$.
- ☐ (c) A has 3 distinct eigenvalues for all $h \in \mathbb{R}$.
- ☐ (d) A can be diagonalized for all $h \in \mathbb{R}$.

Risposta errata.

The correct answer is: A can be diagonalized for all $h \in \mathbb{R}$.

Question 8

Correct

Mark 100.00 out of 100.00

In the euclidean 2-dimensional space with a fixed cartesian system, consider the conic $\mathcal{C}$ having equation

$$x^2 - 6xy + 9y^2 + 2x - 6y = 0.$$

Which of the following statements is true?

- ☒ (a) $\mathcal{C}$ is degenerate. ✓
- ☐ (b) $\mathcal{C}$ is a not degenerate parabola.
- ☐ (c) $\mathcal{C}$ is a not degenerate hyperbola.
- ☐ (d) $\mathcal{C}$ is a not degenerate ellipse.

Risposta corretta.

The correct answer is: $\mathcal{C}$ is degenerate.

Question 9

Correct

Mark 100.00 out of 100.00

In the euclidean 3-dimensional space with a fixed cartesian system, let us consider the circumferences of equations

$$\mathcal{C}_1 : x^2 + y^2 + z^2 - 1 = z - 1 = 0, \quad \mathcal{C}_2 : x^2 + y^2 + z^2 - 1 = z = 0,$$

having radius respectively R_1 and R_2 .

Which of the following statements is true?

- ☒ (a) $\mathcal{C}_1$ consists of only one point. ✓
- ☐ (b) None of the other answers is correct.
- ☐ (c) $R_1 > R_2$.
- ☐ (d) $R_1 = R_2$.

Risposta corretta.

The correct answer is: $\mathcal{C}_1$ consists of only one point.

Question 10

Incorrect

Mark -15.00 out of
100.00

Let us consider the quadratic form defined by

$$q(x, y, z) = (x, y, z)B \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 3x^2 - 6xy + 4y^2 - 3z^2.$$

Which of the following statements is true?

-
- ☒ (a) $q(a, b, c) \geq 0$ for every $(a, b, c) \in \mathbb{R}^3$.
✗
- ☐ (b) There exists $(a_1, b_1, c_1), (a_2, b_2, c_2) \in \mathbb{R}^3$ such that $q(a_1, b_1, c_1) \cdot q(a_2, b_2, c_2) < 0$.
- ☐ (c) The quadratic form $Q(x, y) = q(x, y, 0)$ is negative definite.
- ☐ (d) $q(a, 0, c) < 0$ for every $(a, c) \in \mathbb{R}^2$.

Risposta errata.

The correct answer is: There exists $(a_1, b_1, c_1), (a_2, b_2, c_2) \in \mathbb{R}^3$ such that $q(a_1, b_1, c_1) \cdot q(a_2, b_2, c_2) < 0$.